

Population density, activity centres, and pandemic: Visualizing clusters of COVID-19 cases in Hong Kong

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Abstract

The impacts of coronavirus disease 2019 (COVID-19) on society and economy are wide-ranging, long-lasting, and global. The experience of multiple countries or regions in fighting the pandemic indicates that there could be multiple COVID-19 surges, where a growing number of cases can be observed in the more recent surge(s). Were COVID-19 cases and clusters of cases (across surges) randomly distributed in spaces? Did population density and activity centres influence clusters of cases and associated venues? Based on information on the associated venues of the four surges of COVID-19 cases between January 2020 and February 2021 as well as population density, visuals were made to distinguish the relationships between population density, activity centres, and clusters of cases in Hong Kong. Different spatial patterns were observed across the four surges: fewer cases were observed in the first surge with a more evenly distributed pattern of clusters; the second surge as compared to the first surge saw a wider distribution and an increase in the number/layer of clusters; compared to the second surge, the third surge suffered from many more cases but saw a decrease in the general number of clusters; and compared to the previous

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three surges, the fourth surge had the largest number of cases, yet even fewer clusters were observed, where several clusters are again concentrated in specific areas similar to the previous surge. Across the four surges, a few locales could see recurrent clusters of cases and a few communities were without cases.

Keywords

COVID-19, footprint, population density, activity centres, Hong Kong

The impacts of coronavirus disease 2019 (COVID-19) on society and economy are wide-ranging, long-lasting, and global. The experience of multiple countries or regions in fighting the pandemic indicates that there could be multiple surges of pandemic outbreaks, where more cases can be observed in more recent surges (Ting and Chueng, 2020). Were those cases and clusters of cases in different surges randomly distributed in spaces? Did population density and activity centres influence the case (cluster) distribution? Decision-makers cannot easily circumvent these questions, which must be addressed effectively to best contain the spread of the pandemic. Based on information on buildings or locales associated with the four officially designated COVID-19 surges between January 2020 and February 2021 as well as the population density by the Tertiary Planning Unit in Hong Kong, visuals were produced to answer the above-raised questions (see Figure 1).

The visuals show that the four surges display different spatial patterns, i.e. the number, distribution, and footprint clusters of the confirmed cases. Most notably, the first surge saw the fewest cases, where the clusters identified using kernel density were evenly distributed across multiple locales in Hong Kong. Most clusters seemed to be in activity centres such as Central and Tsim Sha Tsui across the Victoria Harbour. New Towns (i.e. Tuen Mun, Ma On Shan, and Tai Po) in New Territories West, East, and North, which occupy a large share of Hong Kong's land and house millions of local residents, interestingly, saw much fewer cases and a lower density of clusters as compared to elsewhere.

In the second surge, an overall increase in the distribution and the clusters can be observed as compared to the first surge. The local activity centres across the Victoria Harbour again were 'hotspots' for the clusters. Interestingly, many of the most densely populated areas still did not observe any clusters of cases. This is in line with Hamidi et al. (2020), who argued that population density did not aggravate COVID-19 cases across counties in the US. In other words, population density also tended not to be associated with the COVID-19 'hotspots' as well as their occurrence and diffusion in Hong Kong.

Compared to the first and second surges, the third surge suffered from many more cases and saw a wider spread of clusters. Unfortunately, the local activity centres across Victoria Harbour were still the 'hotspots'. Furthermore, suburban centres such as Mong Kok and urban fringes such as Tsz Wan Shan emerged as new 'hotspots'.

Compared to the previous three surges, the fourth surge had the most cases yet fewer notable clusters, jamming in the densely populated suburban centres, especially around the Yau Tsim Mong area. A few suburban locales such as Sham Shui Po and Mong Kok saw the reoccurrence of clusters. These locales with a high concentration of COVID-19 clusters are among the earliest planned areas in Hong Kong, with a dense grid shape urban form corresponding to a high population density (Zhang, 2000). However, New Towns in the New Territories were planned in a compact yet comprehensive manner, where a relatively sparse urban form as compared to the old districts and a high population density can also be observed (Lang et al., 2019).

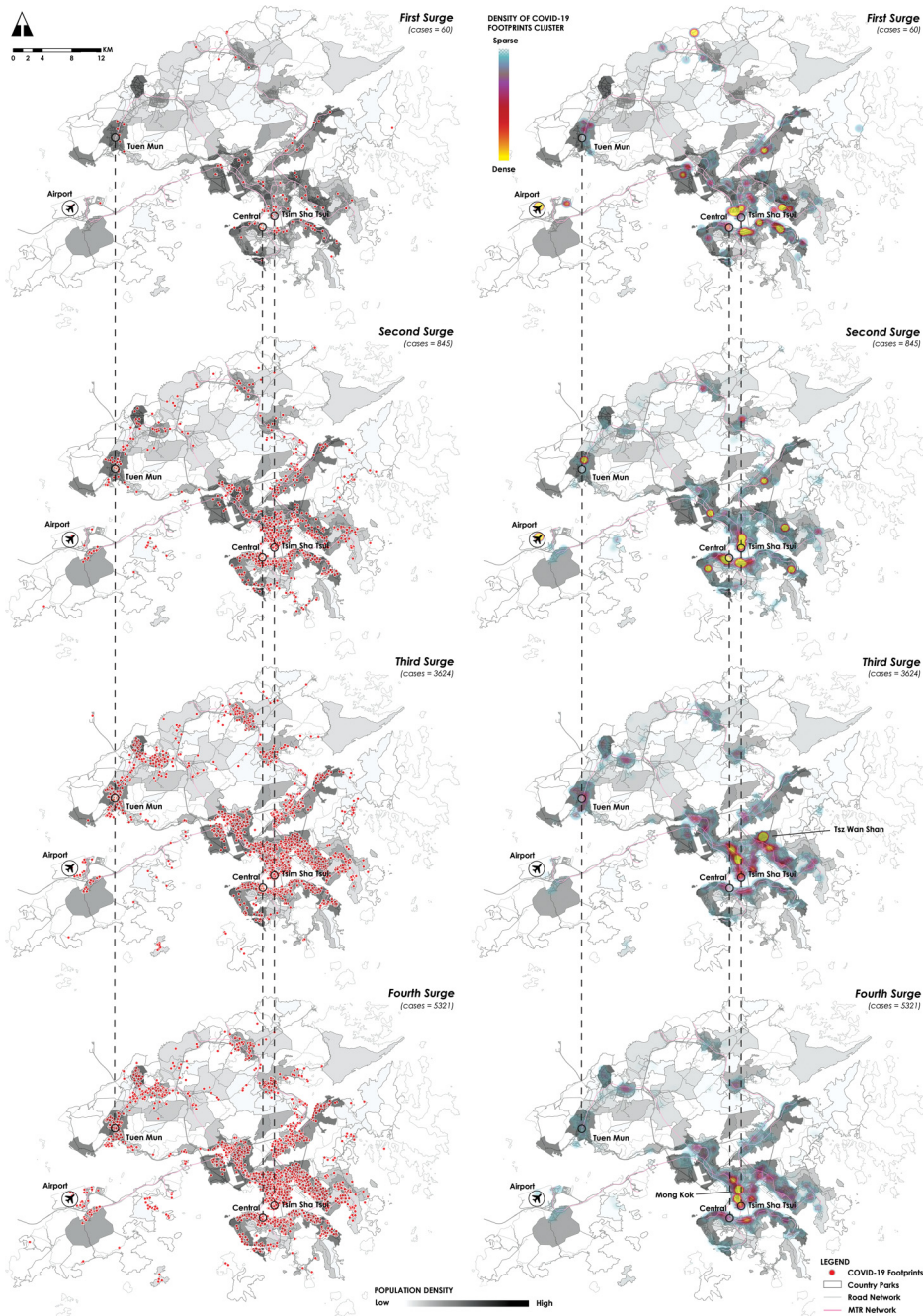


Figure 1. The density of COVID-19 footprints cluster in Hong Kong.
Notes: According to the local government's COVID-19 dashboard: first surge, 23 January 2020 and 17 February 2020, when the city saw the earliest batch of locally confirmed cases; second surge, 16 March 2020 and 11 April 2020; third surge, 1 July 2020 and 31 August 2020; and, fourth surge, 19 November 2020 and 17 February 2021, when the city suffered from the largest number of confirmed cases across the four surges – the number of confirmed cases in the fourth surge was even larger than the combined total of cases in the previous three surges.

The above findings based on the visuals could reflect that (1) without stringent and effective mitigation/preventive measures, there could be more COVID-19 cases yet fewer clusters when the second, third and fourth COVID-19 outbreaks hit a city, (2) the spatial distribution of the clusters was not random – several activity centres were identified to have comparatively more reoccurrence of clusters than elsewhere, (3) a study into the relationships between population density, urban form, and COVID-19 incidences should be explored, and (4) to better understand and contain the COVID-19 pandemic, more spatiotemporal data should be collected and more in-depth analyses of those data should be done (cf. European Commission, 2020).

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